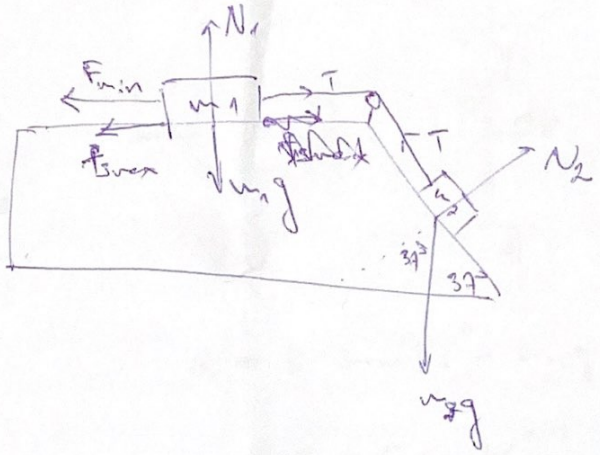


$f_{s\,max} = 21\ N$

$w_2 g \sin 37° = 24.07\ N$

$T = w_2 g \sin 37°$ (1)

$F + f_{s\,max} = T$

$F + \mu_s N_1 = T$

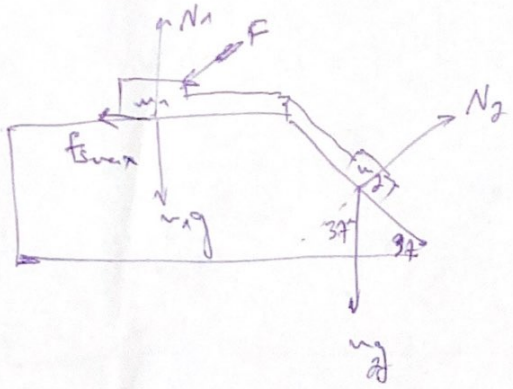
$F + \mu_s w_1 g = w_2 g \sin 37°$

$F = -0.7 \cdot 3 \cdot 10 + 4 \cdot 10 \cdot 0.6018$

$F = 4.07\ N$

$F_{min} = 24.07 - 21$

$F_{min} = 3.07\ N$



$N_1 - F \cdot \sin 45^\circ - m_1 g = 0$

$N_1 = m_1 g + F \sin 45^\circ$

$f_{s\,max} = \mu_s N_1$

$N_1 = 3.10 + 0.707F$

$\boxed{N_1 = 30 + 0.707F}$

$f_{s\,max} = \mu_s N_1$ ⟵

$f_{s\,max} = 0.7 N_1$

$\boxed{f_{s\,max} = 21 + 0.495F}$

$\boxed{m_2 g \sin 37^\circ = 24.07\ N}$

F_{min}

$F_{min} \cdot \cos 45^\circ + f_{s\,max} = T$

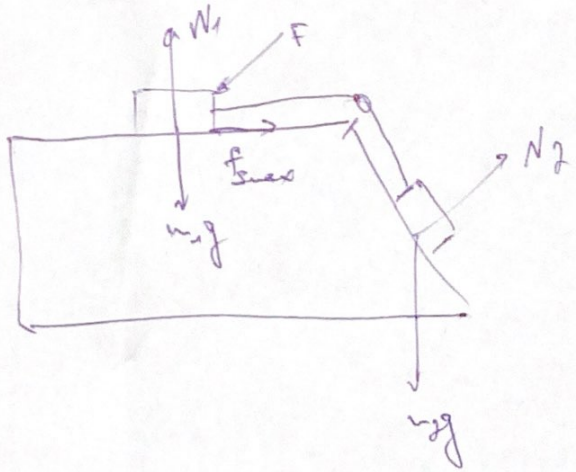
$0.707F + 21 + 0.495F = m_2 g \sin 37^\circ$

$0.707F + 21 + 0.495F = 24.07$

$1.202F = 3.07$

$\boxed{F_{min} = 2.554\ N}$

(c)



$$f_{s\,max} = \mu_s N_1$$

$$f_{s\,max} = 21 + 0.495F$$

$T = m_2 g \sin 37°$

$T = 24.07\ N$

$$0.707F = f_{s\,max} + T$$

$$0.707F = 21 + 0.495F + 24.07$$

$$0.212F = 45.07$$

$$F_{max} = 212.594\ N$$